Algebra Review: Unit 2 — Inequalities

Station 1 of 4: One- and Two-Step Inequalities

Name: _____ Date: _____

Solve each inequality and write the answer as an inequality (for example, $x > 3$). Remember to flip the sign when multiplying or dividing by a negative.

1. Solve: $x + 6 > 10$

2. Solve: $y - 5 \leq 3$

3. Solve: $4n \geq -20$

4. Solve: $-3m < 12$

5. Solve: $\dfrac{x}{-2} \leq 5$

6. Solve: $2x + 5 < 17$

7. Solve: $7 - 3y \geq 22$

8. Solve: $\dfrac{n}{4} - 3 > 1$

9. Solve: $-5x - 2 \leq 18$

10. A gym charges a \$25 sign-up fee plus \$15 per month. Marcus can spend at most \$100. Write and solve an inequality for the number of months m he can join.

Algebra Review: Unit 2 — Inequalities

Station 2 of 4: Multi-Step Inequalities

Name: _____ Date: _____

Solve each inequality. If every real number is a solution or there is no solution, say so.

1. Solve: $3(x-2) > 12$

2. Solve: $5x - 4 \leq 2x + 11$

3. Solve: $6 - 4x < 2x + 24$

4. Solve: $-2(3y - 1) \geq 4y + 22$

5. Solve: $\dfrac{x+3}{2} \leq x - 1$

6. Solve: $\dfrac{2x}{3} - 1 > 5$

7. Solve: $4(x+1) - 3x < 2x + 7$

8. Solve: $0.4x + 3 \geq 0.1x + 4.5$

9. Solve: $3(2x - 1) \leq 6x + 5$

10. Solve: $2(x+4) > 2x + 9$

Algebra Review: Unit 2 — Inequalities

Station 3 of 4: Compound Inequalities

Name: _____ Date: _____

Solve each compound inequality. Write the answer using inequality symbols.

1. $-3 < x + 2 < 7$

2. $4 \leq 2x \leq 10$

3. $-6 < 3x - 3 \leq 9$

4. $1 \leq \dfrac{x+1}{2} < 4$

5. $x - 4 < -3$ or $x - 4 > 3$

6. $2x + 1 \leq -5$ or $3x - 2 \geq 7$

7. $-2x > 6$ and $x + 5 > 0$

8. $3x - 1 < 8$ or $5x \geq 40$

9. Write a compound inequality for all real numbers that are at least -2 and less than 6.

10. Write each in interval notation: (a) $-1 < x \leq 4$ (b) $x < 1$ or $x > 7$

Algebra Review: Unit 2 — Inequalities

Station 4 of 4: Absolute Value Equations and Inequalities

Name: _____ Date: _____

Solve each equation or inequality. Write inequality answers using inequality symbols.

1. $|x| = 7$

2. $|x - 3| = 5$

3. $|2x + 1| = 9$

4. $3|x - 2| = 12$

5. $|x + 4| = -2$

6. $|x| < 4$

7. $|x - 1| \geq 5$

8. $|2x - 3| \leq 7$

9. $|x + 2| > 3$

10. A bolt must be 5 cm long, with a tolerance of 0.02 cm. Write an absolute value inequality for the acceptable length L, then write it as a compound inequality.

Algebra Review: Unit 2 — Inequalities

Answer Key

Station 1: One- and Two-Step Inequalities

1. $x > 4$
2. $y \leq 8$
3. $n \geq -5$
4. $m > -4$
5. $x \geq -10$
6. $x < 6$
7. $y \leq -5$
8. $n > 16$
9. $x \geq -4$
10. $25 + 15m \leq 100;\ m \leq 5$

Station 2: Multi-Step Inequalities

1. $x > 6$
2. $x \leq 5$
3. $x > -3$
4. $y \leq -2$
5. $x \geq 5$
6. $x > 9$
7. $x > -3$
8. $x \geq 5$
9. all real numbers
10. no solution

Station 3: Compound Inequalities

1. $-5 < x < 5$
2. $2 \leq x \leq 5$
3. $-1 < x \leq 4$
4. $1 \leq x < 7$
5. $x < 1$ or $x > 7$
6. $x \leq -3$ or $x \geq 3$
7. $-5 < x < -3$
8. $x < 3$ or $x \geq 8$
9. $-2 \leq x < 6$
10. (a) $(-1, 4]$ (b) $(-\infty, 1) \cup (7, \infty)$

Station 4: Absolute Value Equations and Inequalities

1. $x = 7,\ x = -7$
2. $x = 8,\ x = -2$
3. $x = 4,\ x = -5$
4. $x = 6,\ x = -2$ $\quad |x - 2| = 4$
5. no solution *an absolute value is never negative*
6. $-4 < x < 4$
7. $x \leq -4$ or $x \geq 6$
8. $-2 \leq x \leq 5$
9. $x < -5$ or $x > 1$
10. $|L - 5| \leq 0.02;\ 4.98 \leq L \leq 5.02$